

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

G Hema chowdary

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

G Hema chowdary

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

ARTICLE REVIEW – SOLUTION

Course: Artificial Intelligence (CSA17) **CO4**

Student Name: G Hema chowdary

Selected Research Article

J. Velasco-Mata, V. González-Castro, E. Fidalgo Fernández, and E. Alegre, "Efficient Detection of Botnet Traffic by Features Selection and Decision Trees," *IEEE Access*, vol. 9, pp. 120567–120579, 2021. DOI: 10.1109/ACCESS.2021.3108222.

The article is a 2021 IEEE Access research paper and therefore satisfies the assessment requirement of selecting a peer-reviewed article published after 2019. It studies machine-learning-based botnet traffic classification using feature selection and Decision Trees. ■cite-style■

Article link: <https://doi.org/10.1109/ACCESS.2021.3108222>

Task 1: Article Summary

(a) Citation, domain and ML problem

Citation: Velasco-Mata, Javier; González-Castro, Víctor; Fidalgo Fernández, Eduardo; Alegre, Enrique. "Efficient Detection of Botnet Traffic by Features Selection and Decision Trees." IEEE Access, Volume 9, 2021, pp. 120567–120579. DOI: 10.1109/ACCESS.2021.3108222.

Domain: Cybersecurity and network traffic analysis. **ML problem:** multi-class classification of network traffic into normal traffic and different botnet classes, while reducing the number of traffic features and classification time.

(b) Summary (about 180 words)

The paper addresses efficient botnet traffic detection using supervised machine learning. The authors identify two practical issues in existing approaches: the difficulty caused by severe class imbalance in the CTU-13 dataset and the lack of attention to the computational cost of trained detectors during classification. To address these issues, they construct two datasets, QB-CTU13 and EQB-CTU13, from CTU-13 and additional botnet traffic. They begin with eleven commonly used traffic features and rank their importance using Gini Importance and Information Gain. Based on these rankings, they create feature subsets containing five, six and seven features. The subsets are evaluated with Decision Tree (DT), Random Forest (RF) and k-Nearest Neighbors (k-NN). The study evaluates the trade-off between weighted F1 score and classification time. The main finding is that a Decision Tree using five selected features gives a strong balance of detection performance and speed, with a mean F1 score of about 85% and an average classification time of 0.78 microseconds per sample. Thus, the work shows that careful feature selection can reduce computational cost without making the detector impractical.

Task 2: Methodology Analysis

(a) ML techniques and dataset

Feature selection: The authors start from eleven network-traffic features and rank them using Gini Importance and Information Gain. The selected subsets contain 5, 6 and 7 features. The five-feature subset is: dPort, nPackets, nBytes, vLen and mLen. The six-feature subset adds mTime, and the seven-feature subset further adds vTime.

Decision Tree: A supervised tree-based classifier that recursively separates samples using feature-based decisions. **Random Forest:** an ensemble of Decision Trees whose outputs are combined by voting. **k-NN:** classifies a sample using nearby training samples in feature space.

Datasets: QB-CTU13 is derived from CTU-13 to reduce the effect of class imbalance while retaining under-represented classes and improving the normal class. EQB-CTU13 extends QB-CTU13 with Bunitu, Miuref and NotPetya traffic from Stratosphere IPS.

TCP-flow samples used in the two datasets

Class	QB-CTU13	EQB-CTU13
Normal	3890	3890
Neris	3890	3890
Rbot	3890	3890
Virut	3890	3890
Murlo	2036	2036
NSIS	355	355
Donbot	233	233
Sogou	36	36
Bunitu	–	3890
Miuref	–	3890

NotPetya	–	111
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(b) Mapping to CO4: The work directly demonstrates Decision Trees and Inductive Learning because the classifier learns decision rules from labelled examples. The feature-ranking stage also connects to statistical/feature-selection ideas used in machine learning.

Methodological strength: The study evaluates both predictive performance and computational time, making the analysis more useful for real-time cybersecurity deployment.

Limitation: The experiments are based on constructed datasets derived mainly from CTU-13 and selected additional botnets. Performance may therefore not transfer directly to completely different networks, protocols or unseen attack families.

Task 3: Results & Critical Evaluation

(a) Key results

The authors report that the best overall performance is obtained with a Decision Tree using the five-feature subset. Its mean F1 score is approximately 85%, while the average time required to classify a sample is about 0.78 microseconds. The paper also reports strong class-level results for several botnets. For example, the optimized five-feature Decision Tree achieved 97% recall for NotPetya and 99% F1 and recall for Miuref. For Bunitu, the five-feature model achieved an F1 score of about 84%, compared with 94% when all eleven features were used.

Model / feature setting	Reported finding
Decision Tree + 5 features	Mean F1 $\approx$ 85%; $\approx$ 0.78 μ s/sample
Decision Tree + 5 features – NotPetya	Recall $\approx$ 97%
Decision Tree + 5 features – Miuref	F1 $\approx$ 99%; Recall $\approx$ 99%
Decision Tree + 5 features – Bunitu	F1 $\approx$ 84%
Decision Tree + 11 features – Bunitu	F1 $\approx$ 94%

(b) Critical evaluation

The reported results are encouraging, especially because the authors evaluate F1 together with classification time. However, the very high class-level scores should not automatically be interpreted as equivalent real-world detection performance. The dataset is intentionally constructed to reduce class imbalance, and some classes still have very few samples, particularly Sogou with only 36 TCP flows. This can make class-level metrics unstable. A further concern is dataset shift: network behaviour, encryption, applications and botnet families change over time.

Potential biases: class imbalance, dependence on CTU-13-derived traffic, limited representation of some botnets, and possible differences between laboratory traffic and production networks.

Additional experiments: evaluate on a completely independent modern dataset; test previously unseen botnets; perform time-based train/test splitting; report precision, recall, macro-F1 and confusion matrices for every class; and measure throughput on realistic network hardware.

(c) Real-world applicability

The approach can be deployed in network intrusion detection systems, enterprise gateways, data-centre monitoring, IoT security and SOC pipelines. Its small feature set and very low per-sample classification time are useful for near-real-time monitoring. Industry challenges include collecting labelled traffic, encrypted communications, concept drift, adversarial behaviour, privacy requirements, integration with existing IDS tools and maintaining models as new botnets appear.

Task 4: Reflection & Learning Outcome

(a) Three key insights

1. Feature selection matters: A model does not always need every available feature. Selecting informative variables can reduce computational cost while maintaining useful predictive performance. This connects to **Inductive Learning** and model efficiency.

2. Decision Trees are interpretable: A Decision Tree learns a sequence of feature-based decisions, making the classification process easier to understand than many black-box models. This directly connects to the **Decision Tree** topic in CO4.

3. Evaluation must consider deployment cost: Accuracy/F1 alone is not enough for a real-time cybersecurity system. Classification time is also important. This connects to **Statistical Learning and model evaluation** because performance metrics must be interpreted in the context of the application.

(b) Proposed extension

A useful extension would be to test the five-feature Decision Tree on a newer, independent network-security dataset containing unseen botnet families and traffic captured at different times. The experiment could compare the original Decision Tree with Random Forest and an online/incremental learner under a time-based evaluation. The study should report macro-F1, per-class recall, false-positive rate and classification throughput. This is feasible because the original method already uses a small number of traffic features, so the additional experiment mainly requires collecting or selecting a suitable independent dataset and repeating the evaluation protocol. The expected impact is a better estimate of how well the model generalizes to changing real-world network conditions.

Conclusion

The reviewed article demonstrates a practical application of CO4 machine-learning concepts to cybersecurity. By combining feature selection with Decision Trees and measuring both F1 score and classification time, the authors show that a compact model can provide a useful balance between detection capability and efficiency. The work is especially valuable as an example of how inductive learning and Decision Tree classification can be adapted for real-time decision-making.

Reference

J. Velasco-Mata, V. González-Castro, E. Fidalgo Fernández, and E. Alegre, "Efficient Detection of Botnet Traffic by Features Selection and Decision Trees," *IEEE Access*, vol. 9, pp. 120567–120579, 2021. DOI: 10.1109/ACCESS.2021.3108222.